



Chapter 3: Planning Construction Materials

3.1 ABC Classification of Materials

a) Definition:

It is concerned with planning, organizing and controlling the flow of materials from their initial purchase through internal operations to the service point through distribution.

OR

Material management is a scientific technique, concerned with Planning, Organizing & Control of flow of materials, from their initial purchase to destination.



Definition (ABC Classification)

- ABC plan is based upon segregation of materials for selection control.
- It measures the money value, i.e., cost significance of each material item in relation to total cost and material value.
- The study of each item of stock in terms of its usage, lead time, technical or other problems and its relative money value in the total investment in inventories.
- Critical, i.e., high value items deserve very close attention, and low value items need to be devoted minimum expense and effort in the task of controlling inventories.



Definition (ABC Classification)

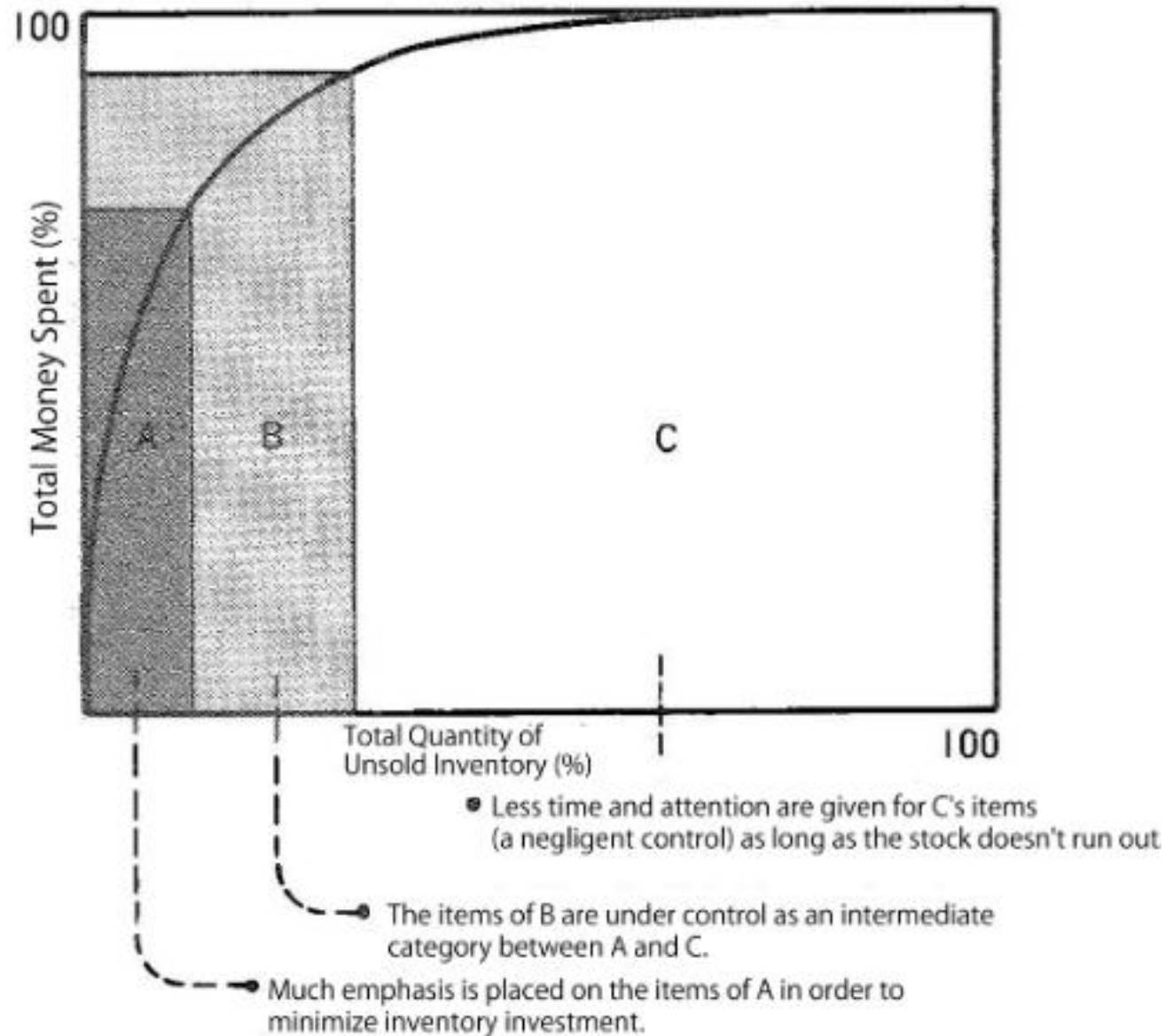
(ABC = Always Better Control)

- This is based on cost criteria.
- It helps to exercise selective control when confronted with large number of items it rationalizes the number of orders, number of items & reduce the inventory.
- About 10 % of materials consume 70 % of resources
- About 20 % of materials consume 20 % of resources
- About 70 % of materials consume 10 % of resources

Group (Class)	Total Items (Stock)	Inventory Cost
A	10 % (5-15 %)	70 %
B	20 % (15-25 %)	20 %
C	70 % (65-75 %)	10 %



Graphical Representation (ABC Classification)





Definition (ABC Classification)

- ABC analysis is an ***inventory categorization method*** which consists in dividing items into three categories (A, B, C):
 - A being the most valuable items,
 - C being the least valuable ones.
- This method aims to draw managers' attention on the critical few (A-items) not on the trivial many (C-items).

"A" Items

Small in number, but consume large amount of resources

About 10 % of materials consume 70 % of resources

Must have:

Tight control

Rigid estimate of requirements

Strict & closer watch

Low safety stocks

Managed by top management



Definition (ABC) Classification

“B” Items

Intermediate.

About 20 % of materials consume 20 % of resources

Must have:

Moderate control

Purchase based on rigid requirements

Reasonably strict watch & control

Moderate safety stocks

Managed by middle level management

“C” Items

Larger in number, but consume lesser amount of resources

About 70 % of materials consume 10 % of resources

Must have:

Ordinary control measures

Purchase based on usage estimates

High safety stocks

ABC analysis does not stress on items those are less costly but may be vital



Essential Features (ABC Classification)

Category A

It involves Precise and Exact Estimation of Inventory Value.

The senior managers and professionals are involved in managing the inventory.

It requires very strict degree of control.

Category B

It involves an approximate estimation of inventory value.

Mid-level professionals and managers are assigned to manage this category of inventory.

It requires moderate degree of control.

Category C

It requires minimal or no estimation of inventory value.

This category is usually managed by the junior staff as it does not require much experience.

It requires minimal or limited degree of control.



ABC Classification

Advantages and Disadvantages:

Advantages

- Facilitates inventory control and control over-usage
- Eliminates unnecessary paperwork involved in control procedures.
- Facilitates selective control thereby freeing up management time
- Reduces stock-holding costs.
- The company is able to concentrate on high value items.

Disadvantages

- Items of low value, but important may be ignored.
- Does not permit precise consideration of all relevant inventory management problems e.g. adequate handling.
- The use of volume is a wrong performance measure; instead, some measure of profitability is more desirable.
- It may need the accumulation of a lot information.



Control Technique (ABC Classification)

ABC Inventory Classification: Materials Management involves thousand of individual transactions each year. To do their job effectively, materials managers must avoid distraction of unimportant details and concentrate on significant matters.

Inventory control procedures should isolate those items that require precise control from those that do not. SIC can indicate where the Manager should concentrate his efforts.

Physical Inventory control still remains a problem in inventory management and the ABC control concept, when applied in this area (to inspection, storage, frequency of cycle checks, etc.) still remain valid.

ABC analysis leads to classification of Inventory items on the basis of their USAGE in monetary terms.

- A – High consumption value items .
- B – Moderate consumption value items.
- C – Low consumption value items.



Use of ABC Analysis for Developing Policy Guidelines.

<i>Nature of Control Procedures</i>	<i>A Items: High Consumption Value</i>	<i>B Items: Moderate Consumption Value</i>	<i>C Items: Low Consumption value</i>
1. Type of control and authority	Very strict control. The controller should have great authority	Moderate control. Controller may be from the middle management.	Low control. Powers can be delegated to the user departments to determine stock level.
2. Quantity of safety stock	Very low or practically nil safety stock, combined with frequent ordering and/or staggered supplies.	Low safety stock. Ordering can be done monthly or quarterly.	High safety stock and bulk ordering. Half yearly or annual orders to take advantage of bulk discounts.
3. Consumption control	Consumption control and flow should be regular—weekly or daily.	This period can be extended to a fortnight or a month.	This period can be extended to a quarter.
4. Material planning	Material planning should be accurate. Data base should be up-to-date.	Past consumption can be used as a basis.	Rough estimates are sufficient.
5. Application of value analysis	A concerted attempt should be made at value analysis, waste reduction, obsolete and surplus reduction.	Moderate attempts are sufficient.	Annual reviews are sufficient.
6. Number of sources of supply	Increase the number of sources. Centralise purchase and stores. Reduce lead time.	2 or 4 reliable sources. Combined purchase. Moderate attempt to reduce lead time.	1 or 2 reliable sources. Annual or half-yearly purchase. Decentralised and low clerical effort.



ABC Analysis

Example-1

Explosive Product	Sales (units)	Unit Price (\$)	Dollar Value	Percent Value	ABC Class
PowXergel Emulsion P100	17,575,110 kg	0.53	9,314,808.30	27.30	A
Porous Prills Ammonium Nitrate	16,219,307 kg	0.42	6,812,108.94	19.97	
Snapline 67ms - 6m	827,138	4.97	4,110,875.86	12.05	
Booster 250g	952,365	3.95	3,761,841.75	11.03	
Unidet 500ms - 10m	199,462	7.12	1,420,169.44	4.16	B
Snapline 17ms - 6m	252,822	4.97	1,256,525.34	3.68	
Booster 400g	227,040	4.80	1,089,792.00	3.19	
Unidet 500ms - 12m	147,038	7.31	1,074,847.78	3.15	
Magnum Bustergel 32x200mm	117,000 kg	6.30	73,7100.00	2.16	
Snapline 25ms - 6m	141,117	4.97	701,351.49	2.06	
Cobracord 6g/m	1,310,254 m	0.52	681,332.08	2.00	
Cordtex 10	949,765 m	0.67	636,342.55	1.87	C
Unidet 500ms - 6m	129,284	4.75	614,099.00	1.80	
Snapline 42ms - 6m	102,627	4.97	510,056.19	1.49	
Unidet 500ms - 7.8m	55,706	6.46	359,860.76	1.05	
Unidet 500ms - 13.5m	44,085	7.71	339,895.35	1.00	
Magnum Bustergel 32x560mm	28,000 kg	6.30	176,400.00	0.52	
Capped Durafuse 3m	83,409	2.10	175,158.90	0.51	
Magnum Bustergel 65x560mm	24,100 kg	6.30	151,830.00	0.44	
Magnum Bustergel 45x560mm	18,000 kg	6.30	113,400.00	0.33	
Unidet 500ms - 4m	19,573	3.45	67,526.85	0.20	
Unidet 500ms - 28m	634	13.56	8,597.04	0.03	
Capped Durafuse 2.1m	1,913	1.90	3,634.70	0.01	



ABC Analysis

Example-2

1	Products	Sold last 3 months	Cumulative Share	Class
2	T-Shirt Bob Said	64	19.9%	A
3	T-Shirt Bob Bondurant	47	34.5%	A
4	T-Shirt Rio Haryanto	31	44.1%	A
5	T-Shirt Helmuth Koinigg	29	53.1%	A
6	T-Shirt Ingo Hoffmann	23	60.2%	B
7	T-Shirt Erik Lundgren	17	65.5%	B
8	T-Shirt Pete Lovely	16	70.5%	B
9	T-Shirt Franck Lagorce	16	75.5%	B
10	T-Shirt John Nicholson	15	80.1%	B
11	T-Shirt Vittorio Brambilla	14	84.5%	B
12	T-Shirt Jack Turner	10	87.6%	B
13	T-Shirt Bernd Schneider	9	90.4%	C
14	T-Shirt Giancarlo Baghetti	8	92.9%	C
15	T-Shirt Eugène Chaboud	6	94.7%	C
16	T-Shirt Joe Kelly	5	96.3%	C
17	T-Shirt Jo Gartner	4	97.5%	C
18	T-Shirt Sakon Yamamoto	3	98.4%	C
19	T-Shirt Derek Warwick	2	99.1%	C
20	T-Shirt Rudolf Krause	1	99.4%	C
21	T-Shirt Bobby Grim	1	99.7%	C
22	T-Shirt A. J. Foyt	1	100.0%	C
23	T-Shirt Otto Stuppacher	0	100.0%	C
24	T-Shirt Stefano Modena	0	100.0%	C
25	T-Shirt Jim Clark	0	100.0%	C
26	T-Shirt Bob Drake	0	100.0%	C



3.2 Material Wastage Standards

Definition:

- **Material Waste:** is defined as **any losses** produced by activities that generate **direct or indirect costs** but do not add any value to the product (Lauri, 1999). It is also defined as the **surplus materials** that are generated from construction, renovation and demolition activities (Formoso, 1999).
- **Construction Waste Management:** is the process of developing and implementing a **strategy** or a **plan** to reduce the generated waste to the minimum extent possible by three approaches: **Reducing, Reusing, and Recycling.**



Wastage of Materials

- Material to be provisioned= Theoretical Quantity x (100 + standard wastage %) /100
- Standard wastage of the construction materials depends up on the followings variable
 - Nature of works
 - Types of materials
 - Method of Application/ operation
 - Working environment
 - Handling procedure
 - Expertise of management/ design
 - Procurement
 - Store keeping



Types of Wastage

➤ **Uncontrollable wastage:** During the usage, material wastage **can't be avoided**. So, actual quantity of material will be more than theoretical quantity.

➤ **Controllable wastage:**
If we know the **causes of wastage** of materials then we can control such wastages. Which are illustrated as shown in table.

Sr. No	Factors	Causes
1	Management	Poor supervision
2		Lack of management
3		Lack of waste reduction plan
4		Absence of site waste manager
5	Operation	Rework
6		Weather
7		Accidents
8		Improper worker's skill
9		Equipment malfunction
10	Design	Changes in design
11		Error in contract documents
12	Handling	Improper packaging
13		Storage
14		Cutting
15		Transportation
16	Procurement	Ordering error
17		Supply error
18	Other	Poor quality control
19		Theft/vandalism



Wastage Standard of Construction Materials

Wastage standard for mostly used construction materials are:

No.	Type of Materials	Planned Wastage
1.	Cement	2%
2.	Sand	10%
3.	Aggregate	5%
4.	Concrete structural	2%
5.	Concrete blinding (lean)	10%
6.	Reinforcement steel bars	3%
7.	Reinforcement steel mesh	10%
8.	PVC sheeting	15%
9.	Steel for windows	7%
9.	Timbering in trenches	5%
10.	Stone masonry	5%
11.	Marble lining	20%
12.	Wood for door frames	5 to 7.5%
13.	Wood for shutters	10%
14.	Wood for flooring/walling	5 to 10%
15.	Sheet sheet roofing	2.1/2%
16.	Tile roofing	5%
17.	Floor tiling	2 to 5%
18.	Wall tiling	3%
19.	Pigments (for colours other than natural grey)	5%
20.	Paints	5%



3.3 Material Provisioning Process

To determine type and quantity of the construction material, it requires detailed study of contract documents e. g. BOQ, drawings, pretender estimates and specifications etc.

Following are material provisioning process:

- Study contract documents
- Estimate quantities
- Locate the sources
- Invite the quotations
- Analyze the quotation and pre-qualification of venders
- Submit proposals (technical, financial & client's approval
- Negotiate with vender and finalization
- Monitor order execution
- Inspect good received at site
- Close material supply contract
- Inform account, planning and project manager



Classification of Construction Materials

Material Type	Details	Example
Bulk materials	Materials that are delivered in mass and are deposited in a container.	Sand, Gravel, Topsoil, Cement, Concrete
Bagged materials	Materials delivered in bags for ease of handling and controlled use.	Cement
Palleted materials	Bagged materials that are placed in pallets for delivery	Cement, Doors
Packaged materials	Materials that are packaged together to prevent damage during transportation and deterioration when they are stored.	Pipes, Tiles, Electrical Fitting



Objectives of Material Management

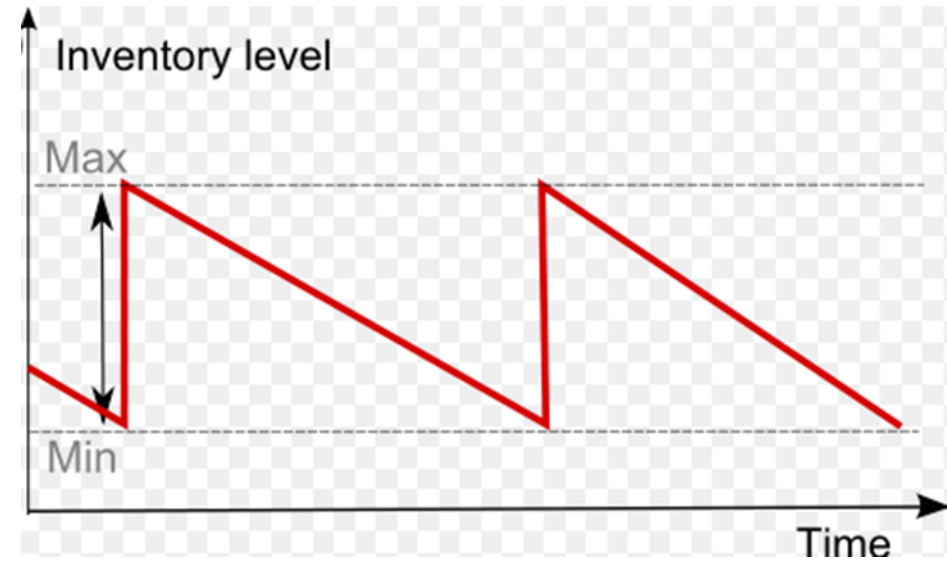
- Efficient production scheduling
- To take make or buy decisions
- Prepare specifications and standardization of materials
- To assist in product design and development
- Forecasting demand and quantity of materials requirements
- Quality control of materials purchased
- Material handling
- Use of value analysis and value engineering
- Developing skills of workers in materials management
- Smooth flow of materials in and out of the organization



3.4 Material Inventory Basics

Definition:

- **Inventory is created** when the receipt of materials, parts, or finished goods exceeds their disbursement.
- **Inventory is depleted** when their disbursement exceeds their receipt.
- An inventory manager's job is to balance the advantages and disadvantages of both low and high inventories.
 - Both have associated cost characteristics.





Material Inventory Basics

Inventory Includes:

- Raw materials
- Work in progress
- Finished goods
- Merchandise (Commodities)
- Spare parts
- Other operating supplies

Inventory may be found in:

- Factories
- Warehouses
- Retail stores
- Other types of storage facilities

- **Inventory** consists of materials, parts, and finished goods. (Stuff in the supply chain, being worked on, and completed but not yet sold.)
- **Inventory is depleted** when it's sold or destroyed.
- An inventory-manager's job is to balance the pros and cons of both low and high inventories.
 - Both low and high inventory levels have associated cost characteristics.
 - The ideal cost situation is to have "some" inventory.



Inventory Management

Definition:

- Inventory management is defined as the long-range, mid-range, and day-to-day processes, protection, and deployment, including the control, of assets in inventory.
- Inventory management strives to balance risk between stockouts, in which no inventory is available to sell, and excess, in which there is leftover inventory.
 - Companies need to have raw material inventory to satisfy their manufacturing needs and finished goods to meet customer needs.
 - At the same time, companies cannot afford to have more inventory than they need because it commits capital dollars and increases the costs of managing inventory



Objectives of Inventory Control

Following are the objective of inventory control:

- To ensure that the supply of raw material & finished goods will remain continuous so that production process is not halted and demands of customers are duly met.
- To minimise carrying cost of inventory.
- To keep investment in inventory at optimum level.
- To reduce the losses of theft, obsolescence & wastage etc.
- To make arrangement for sale of slow moving items.
- To minimise inventory ordering cost.



Importance for Inventory Management

Following are the need / importance on inventory management:

- Lower prices for material and equipment.
- Faster inventory turnover.
- Continuity of supply
- Reduced lead time
- Reduced transportation cost.
- Less duplication of efforts.
- Elimination of bulk- passing.
- Reduced materials obsolescence.
- Improved supplier relationship and better records and information.
- Better inter-department cooperation & Personnel development



Inventory Management

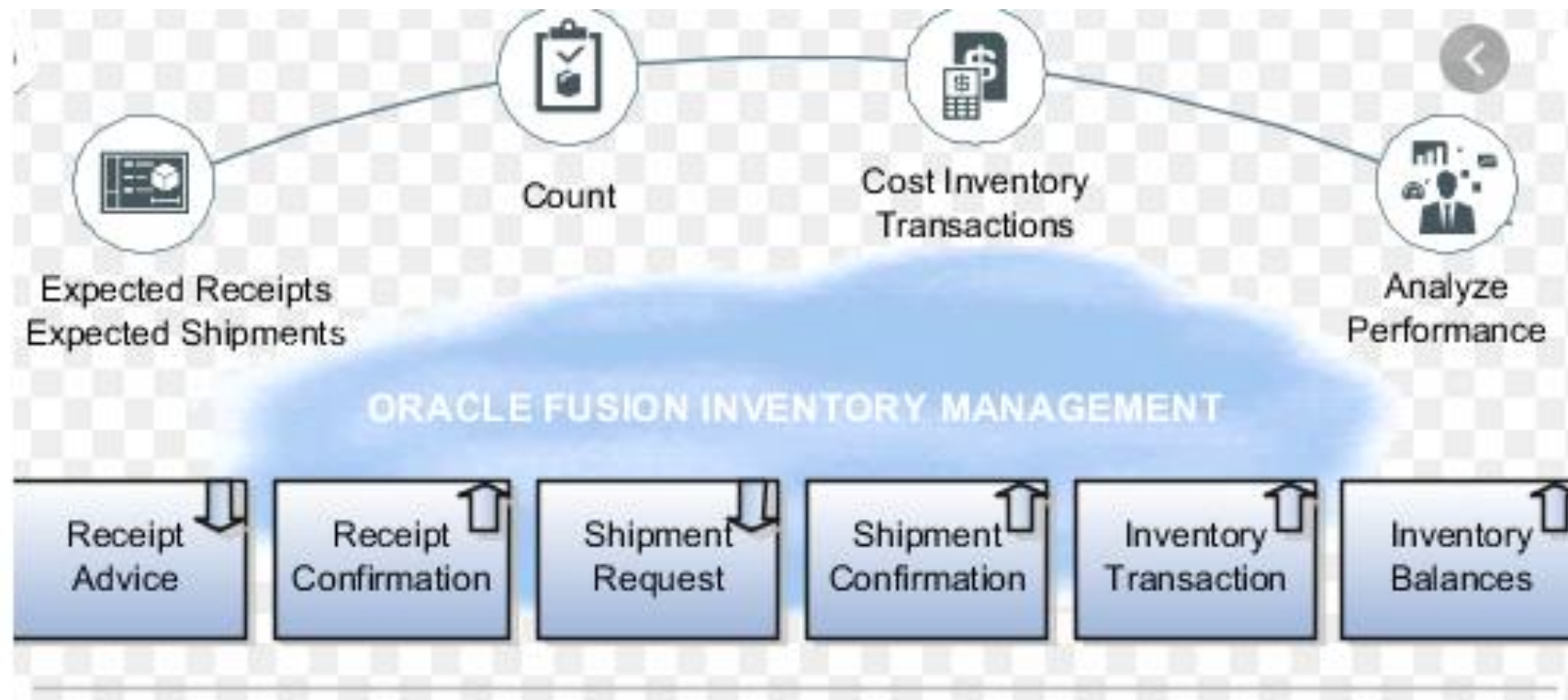
Inventory Management for ABC categories:

No	Inventory Management Actions	A Category	B Category	C Category
1	Management	Top Level	Middle Level	Lower Level
2	Control	Strict	Moderate	Low
3	Review	Very Frequent	Moderate	Routine
4	Consumption Value	High	Medium	Low
5	Purchase	Frequent	Less frequent	Very less frequent



Inventory Management

Steps:





Inventory Control

Tools and Technique-Inventory Control:

- Determination of stock level.
- Determination of safety stock.
- Selecting a proper system of ordering for inventory.
- Determination of EOQ.
- ABC Analysis.
- VED Analysis.
- Inventory turnover ratios.
- Aging schedule of inventories.
- Classification and codification of inventories.
- Preparation of inventory reports.



Inventory Management Cost

A) Purchase Cost:

- ❖ The *purchase cost* of an item is the unit purchase price from an external source including transportation and freight costs.
- ❖ For construction materials, it is common to receive discounts for bulk purchases, so the unit purchase cost declines as quantity increases.
- ❖ Because of this, organizations may consolidate small orders from a number of different projects to capture such bulk discounts, in some cases; this is a basic saving to be derived from a central purchasing office



Inventory Management Cost

B) Order Cost:

- ❖ The *order* cost reflects the administrative expense of issuing a purchase order to an outside supplier.
- ❖ Order costs are usually only a small portion of total costs for material management in construction projects, although ordering may require substantial time.

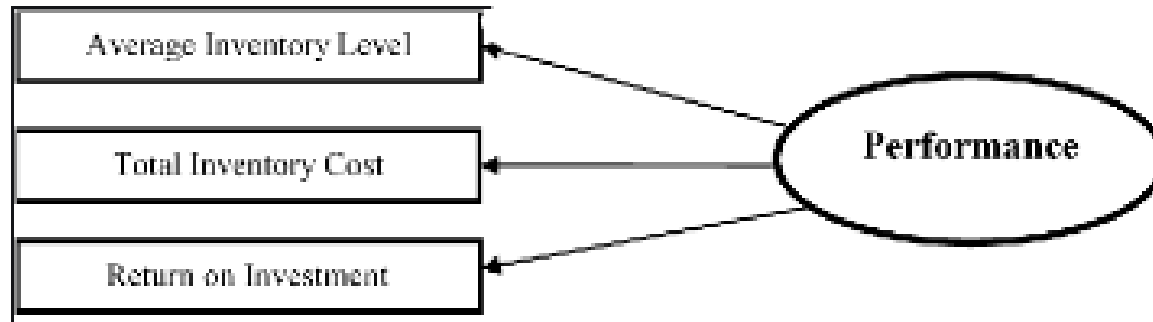
C) Holding Cost:

- ❖ The *holding costs* or *carrying costs* are primarily the result of capital costs, handling, storage, obsolescence, shrinkage and deterioration.
- ❖ Capital cost results from the opportunity cost or financial expense of capital tied up in inventory.
- ❖ Handling and storage represent the movement and protection charges incurred for materials.

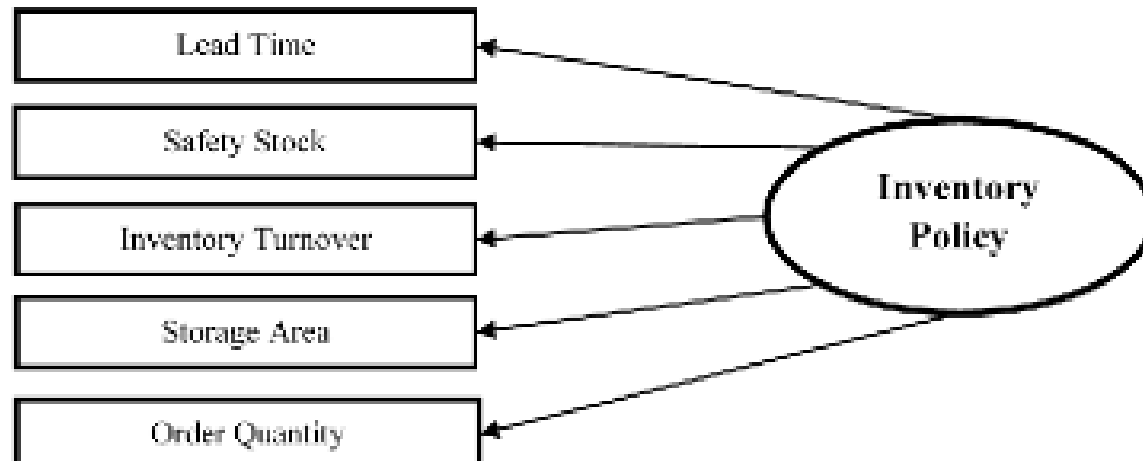


Factors Analysis of Inventory Management

Performance:



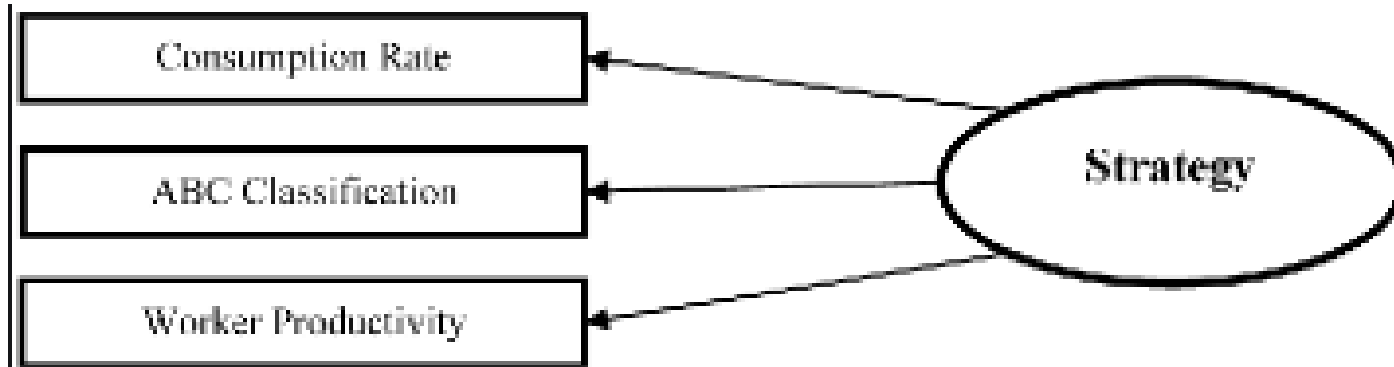
Policy:



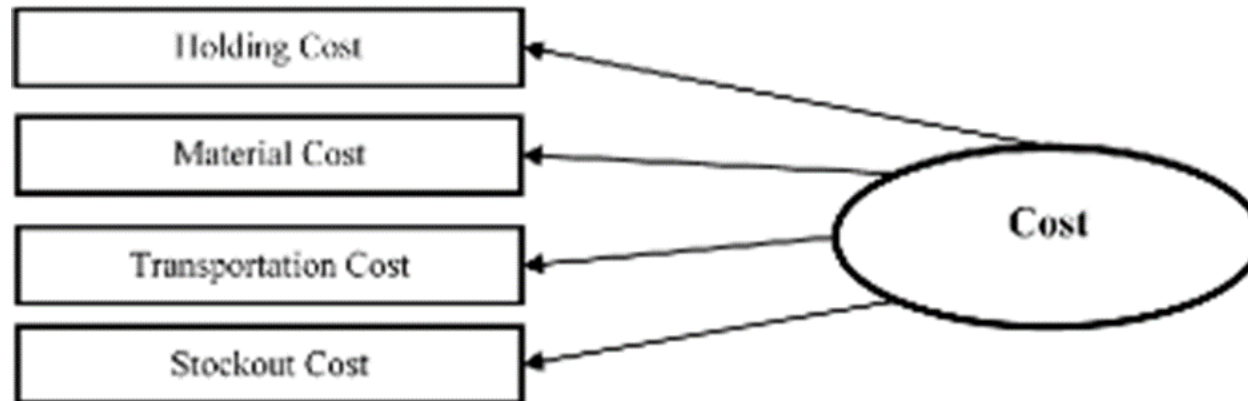


Factors Analysis of Inventory Management

Strategy:



Cost:





Economic Order Quantity (EOQ)

Definition:

- It is a technique of inventory management.
- It determines the **optimum order size** that will minimize the total inventory cost.
- It is a recorder quantity which explains how much quantity has to be ordered at a time to optimize the cost of carrying and holding inventory.
- EOQ is a measurement used in the field of operations, logistics, and supply management.
- In principle, EOQ is **a tool** used to determine the **volume and frequency of orders** required to satisfy a given level of demand while minimizing the cost per order.
- It is a set point designed to help companies minimize the ordering and holding inventory.
- The **cost of ordering** inventory **falls** with the **increase** on ordering **volume** due to purchasing on economics of scale.
- As the size of inventory grows, the cost of holding the inventory rises.
- It is the exact point that minimizes both of these inversely related cost.



Economic Order Quantity (EOQ)

EOQ Graph and Formula:

$$EOQ = \sqrt{\frac{2 \times A \times O}{C}}$$

Where,

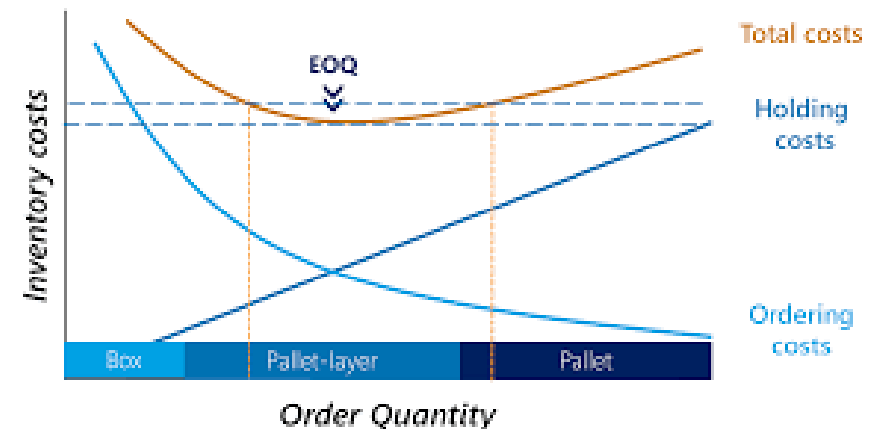
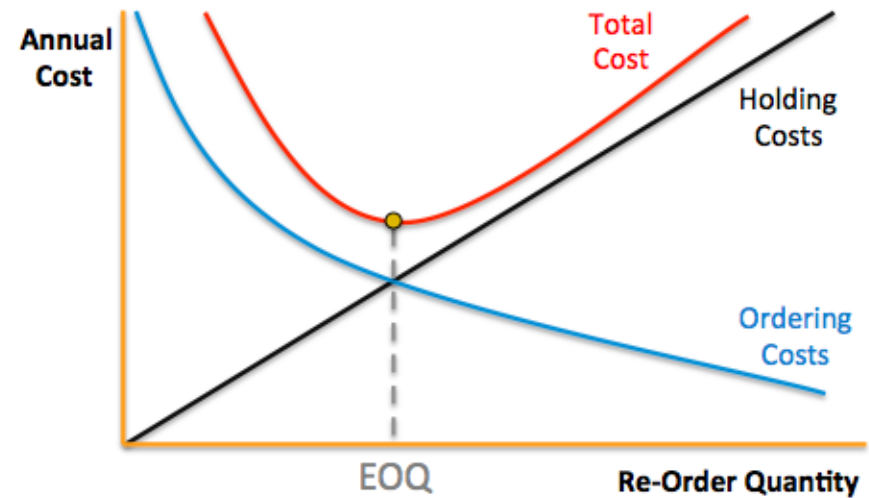
A – Annual material used

O – Ordering cost (Cost to place each order)

C – Carrying Cost

EOQ- Economic Order quantity

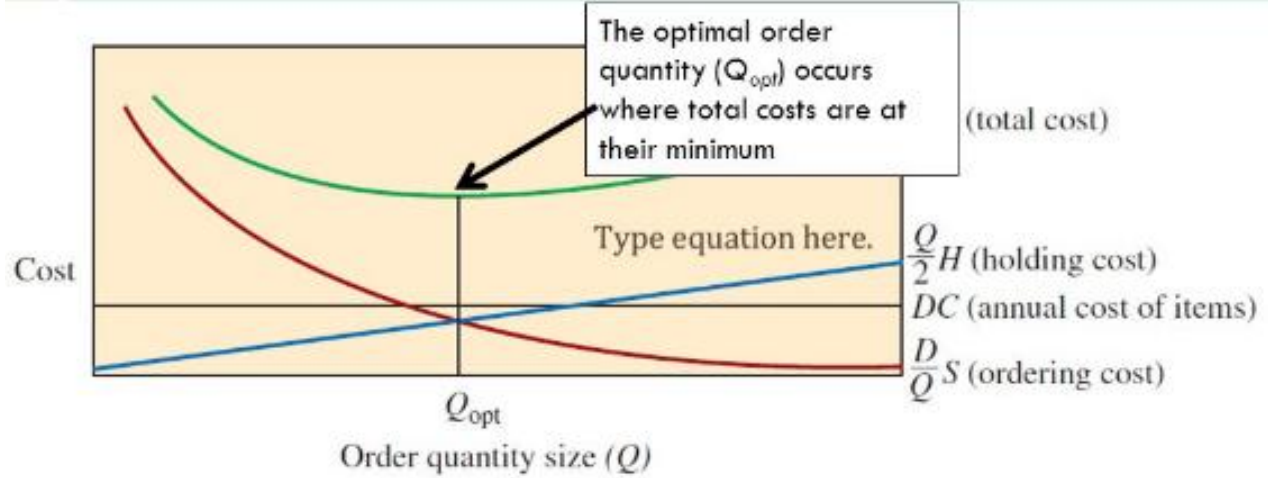
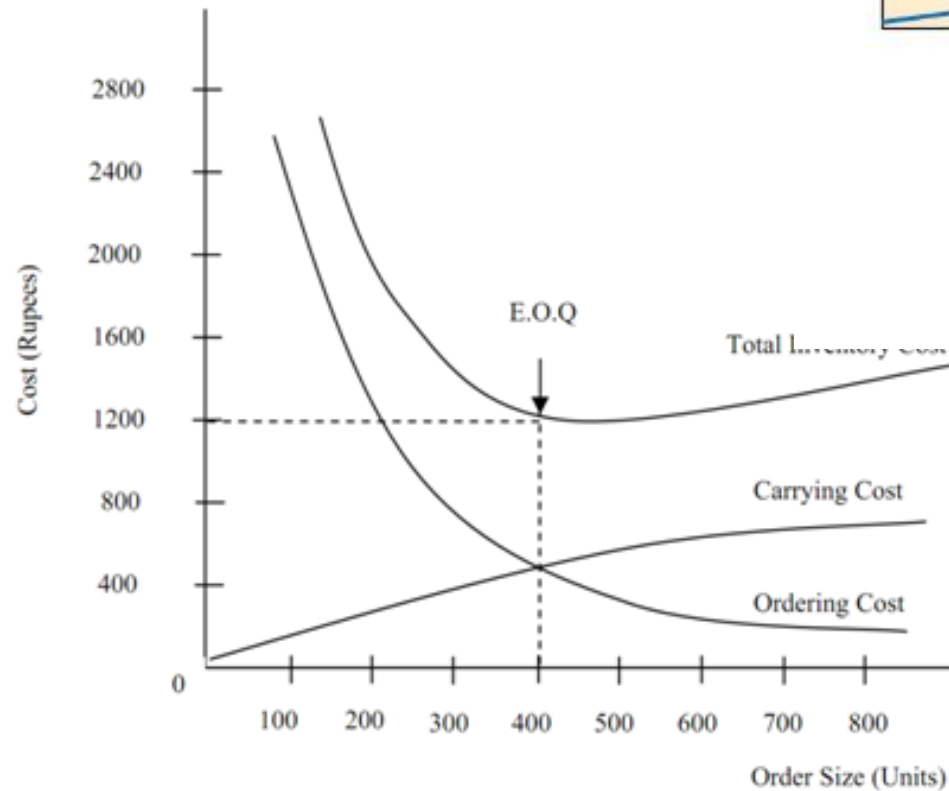
$$\text{Economic Order Quantity (EOQ)} = \sqrt{\frac{2 \times \text{Annual Demand} \times \text{Ordering Cost}}{\text{Inventory Carry Cost per Unit per Year}}}$$





EOQ

EOQ Graph & Formula:



$$Q_{opt} = \sqrt{\frac{2DS}{H}}$$

Carrying costs = ordering costs

$$\frac{Qic}{2} = \frac{AS}{Q}$$

Solving for Q gives

$$Q^2 = \frac{2AS}{ic}$$

$$Q = \sqrt{\frac{2AS}{ic}}$$



EOQ

Economy Lot Size (ELS):

The **Economic Lot Size (ELS)** is a special case of EOQ for manufacturing processes where instead of order costs it is typically considered set-up costs for the manufacturing process.

Definition

$$\text{Economic Lot Size} = \sqrt{\frac{2 * D * S}{K * C}} \quad \text{or} \quad ((2 * D * S) / (K * C))^{\wedge.5}$$

Where :
D = Annual Demand (Historical or Forecasted)
S = Set-up Cost
K = Carrying or Holding Cost Rate (%)
C = Unit Cost

- **Optimal order quantity**
 - Economic order quantity (EOQ)

$$Q^* = \sqrt{\frac{2KD}{h}}$$

- **Important insights**
 - Tradeoff between setup costs and holding costs when determining order quantity. In fact, we order so that these costs are equal per unit time
 - Total cost is not particularly sensitive to the optimal order quantity

Order Quantity	50%	80%	90%	100%	110%	120%	150%	200%
Cost Increase	25.0%	2.5%	0.5%	-	0.4%	1.6%	8.0%	25.0%



EOQ

Formula:

- $EOQ = \sqrt{\frac{2aD}{K}}$

- Where:

a = cost of placing one order
(ordering costs)

D = annual demand in units

K = annual costs of carrying one
unit in inventory for one year

- **Total Inventory Costs** =
Total Ordering Costs + Total Carrying Costs
- **Total Ordering Costs** =
 $\frac{\text{Annual demand in units}}{EOQ} \times \text{Ordering Costs per order}$
- **Total Carrying Costs** =
Average Inventory x Carrying Costs per unit
- **Average Inventory** = $\frac{EOQ}{2}$



Inventory Planning Process

Introduction:

- Each project starts with zero material stock and after completing all activities, it should end up with zero stock.
- It is **ideal material stocking policy** (rarely feasible in construction project).
- **Planning** of inventory is required according to the **usage value** and **type of materials**.

a) Planning Inventory of Repetitive Materials

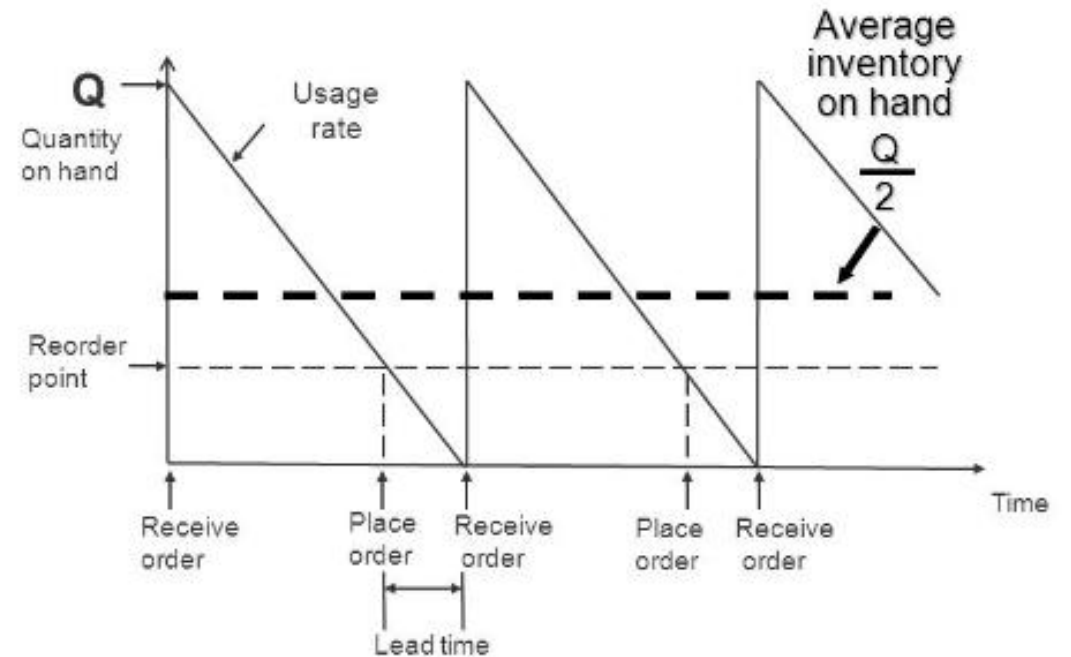
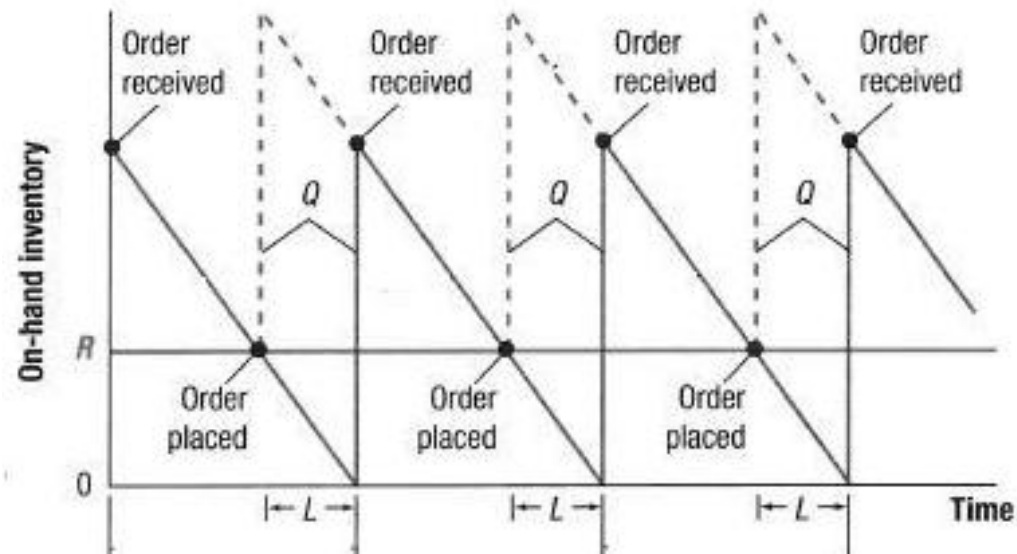
- Determine the economic order quantity
- Fixing the maximum and minimum stock limit
- Lead time for stock refill and reordering point (ROP) for each item of A and B category (high value) materials



Inventory Planning Process

- The simplified model is as shown in figure which contains economic order quantity, working stock, safety stock, maximum & minimum stock level and the number of re-order points (ROP) during the usage period for each item of material.

Profile of Inventory Over Time (Inventory Cycle):





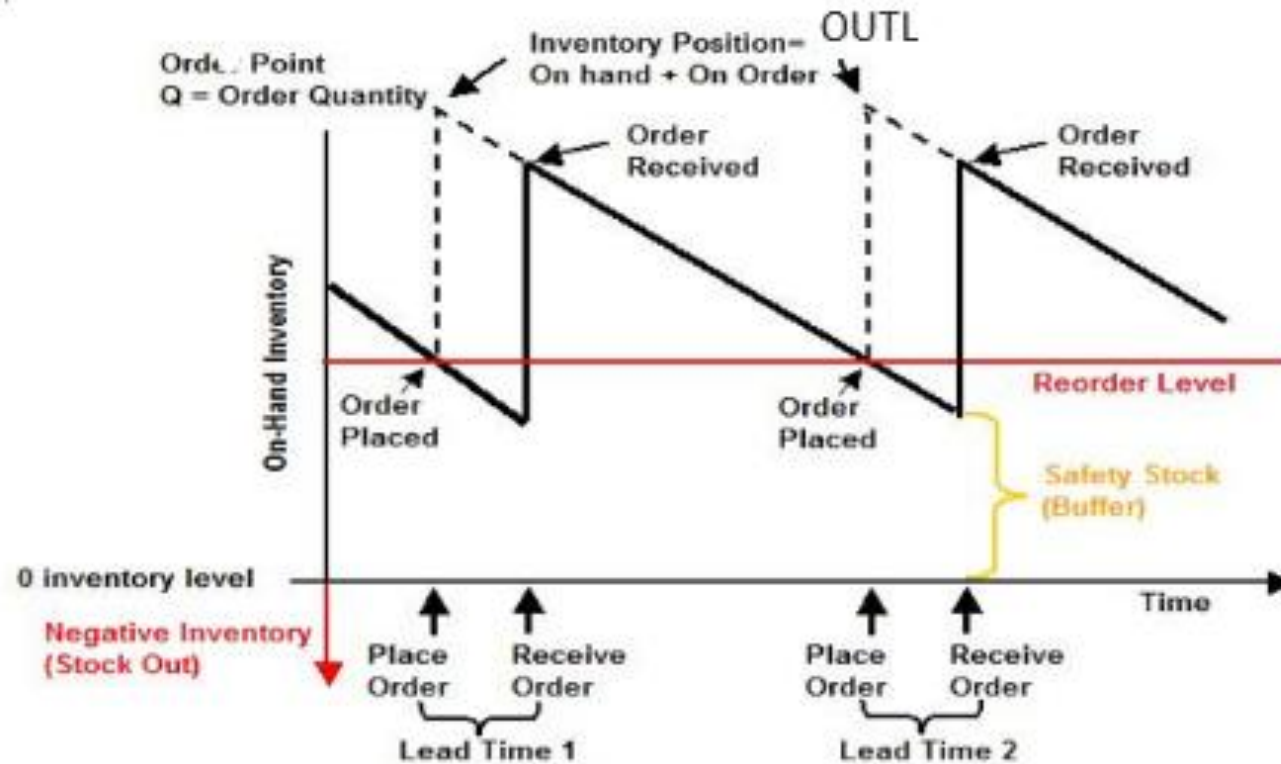
Inventory Planning Process (Terminology)

- **Working Stock:** Normal Consumption quantity during each cycle
- **Work Stock (Q):** Total quantity during period/number of refill cycles
- **Safety Stock:** Floating stock (Minimum stock)
- **Lead time:** Time for order to receipt
- **Maximum stock level:** A line with consumer demand and storage space available. It prevents overstocking
- **Minimum stock level:** A line with normal demand and lead times from suppliers. It prevents understocking
- **Re-order level:** It is set between maximum and minimum level taking account of minimum level, lead time and normal demand. It prevents stock-outs.



Inventory Planning Process

Basic Inventory Model:

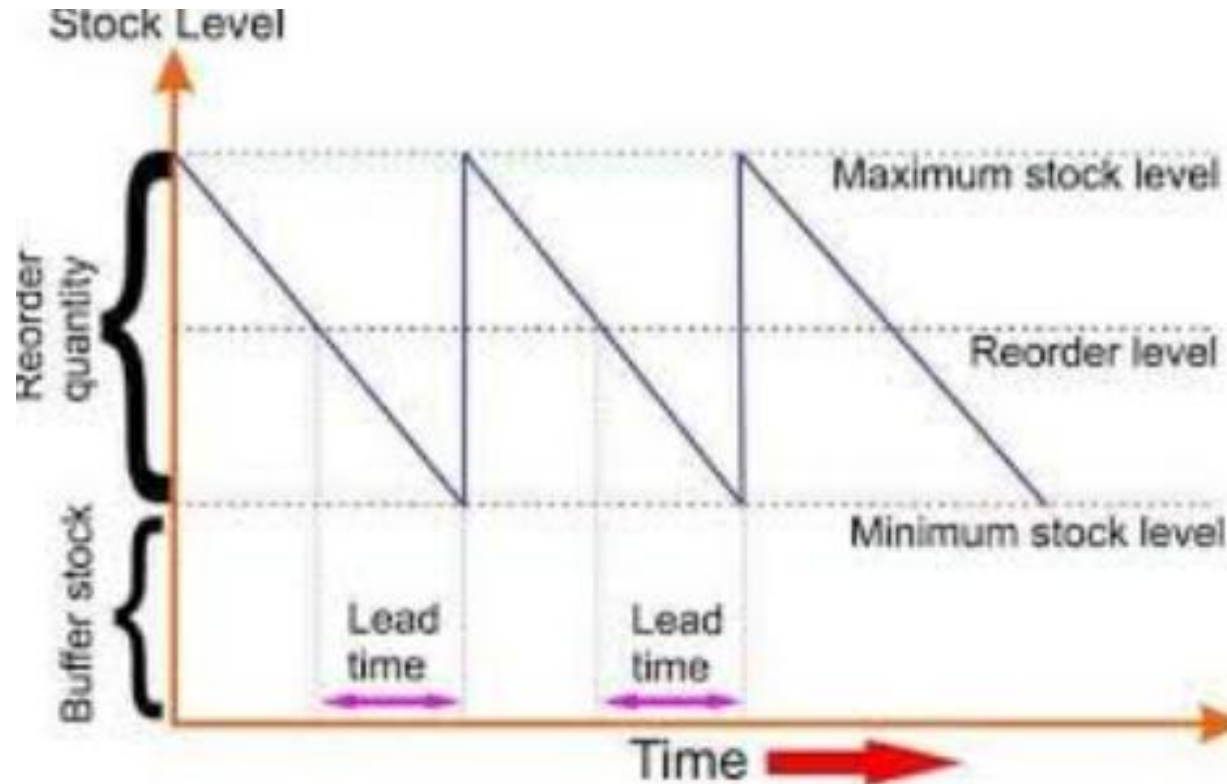


- Reorder level
- Safety Stock
- Lead time
- Order up to Level (OUTL)



Inventory Planning Process

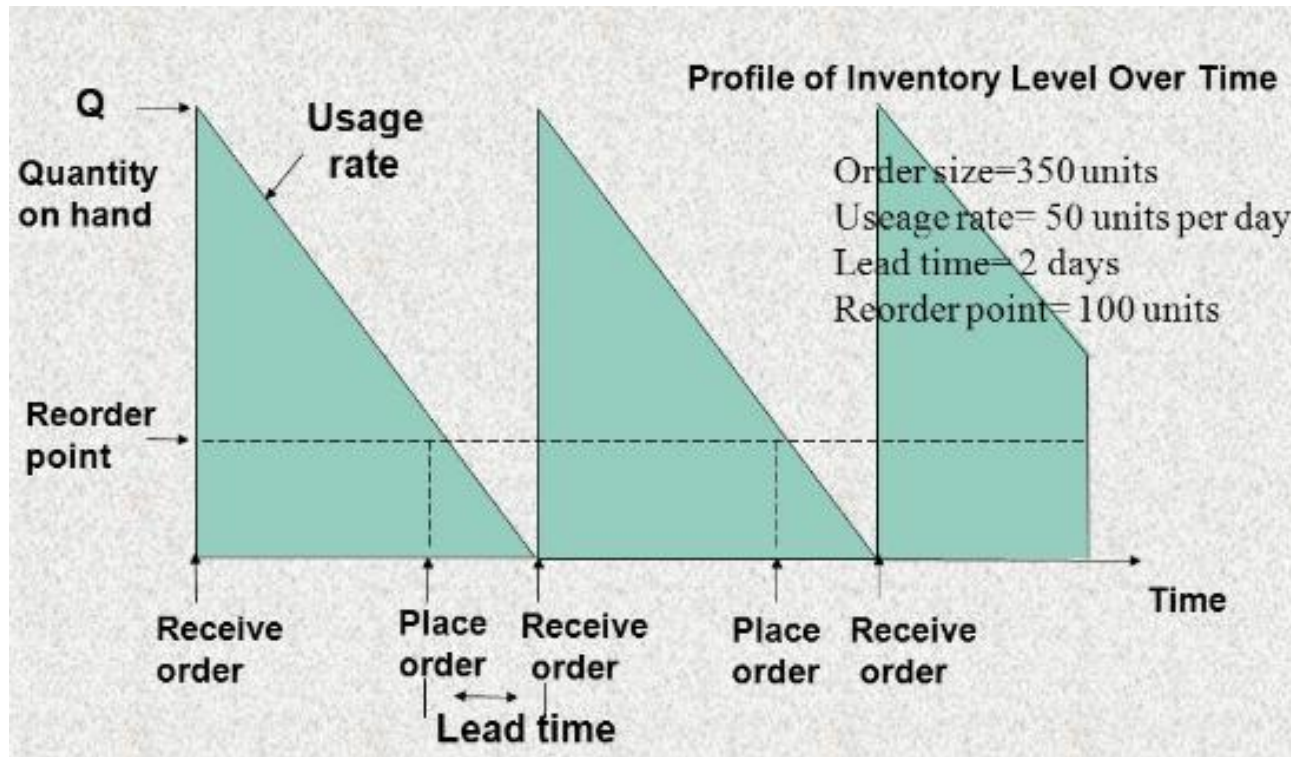
Stock Control:





Inventory Planning Process

Terminology & Graph of Inventory:





Inventory Planning Process

b) Planning Inventory of Non-repetitive (One-Time Purchase) Materials:

- The materials are purchased for specific purpose (e.g. architectural fittings and fixtures, water supply, sanitation, electrical and mechanical works etc.).
- Aim should be to have zero inventory prior to commencement of activity and after completion of job with low average during execution.

c) Minor Material Inventory:

- Low cost, common use and frequently-needed items e.g. screws nails etc.



3.6 Application of Value Engineering in the Procurement of Materials

Value Engineering:

- An **analysis of materials, processes, and products** in which functions are related to cost and from which a selection may be made so as achieve the desired function at the lowest overall cost consistent with performance.
- An **organized, creative, cost search technique** for analyzing the function of a product with the purpose of **value enhancement** without compromising quality, performance & efficiency.

Value= Worth / Cost

- Value means reliable performance
- Function / Worth means natural action performed
- Cost means expenditure



Value Analysis & Value Engineering

- Value analysis & Value engineering is defined as **an organized approach** to provide the required function at the **lowest cost** consistent with specific **performance, quality & reliability**.
- Value analysis refers to the existing product & services where as value engineering is concerned to the design of new Products.

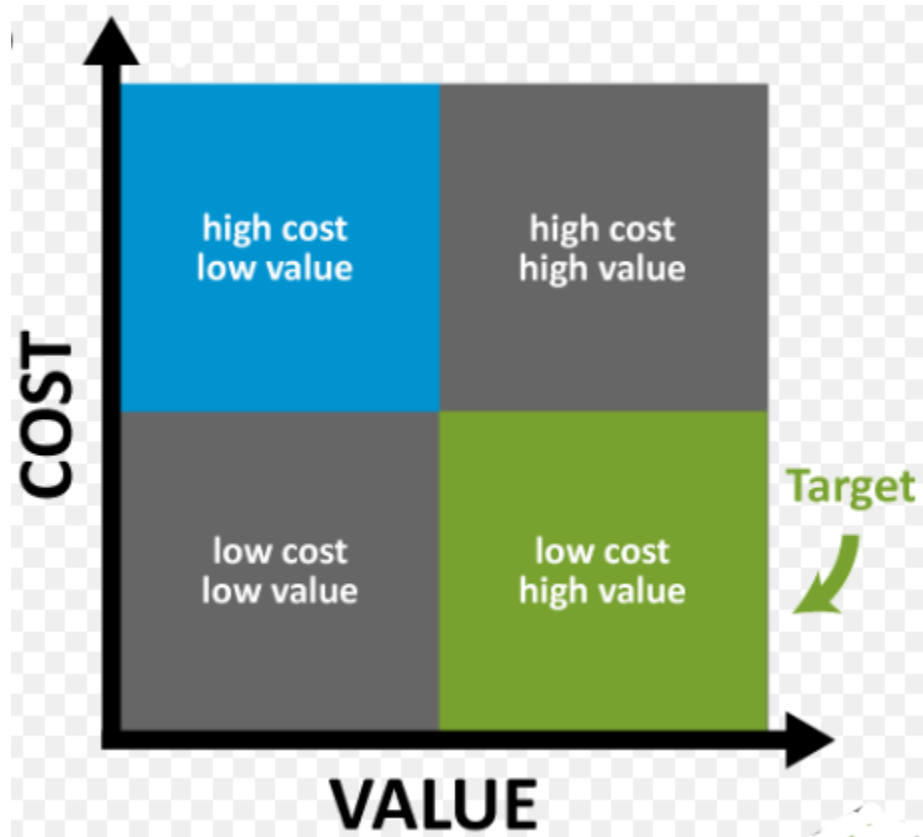
Use of Value Engineering:

- It is used to determine the **best design alternatives** for projects, processes, products, or services
- It is used to **reduce cost**
- It is used **to improve quality, increase reliability & availability, and customer satisfaction**
- It is also used to **improve organizational performance**
- It is **powerful tool** used to identify problems and develop recommended solution



Value Engineering

Definition:

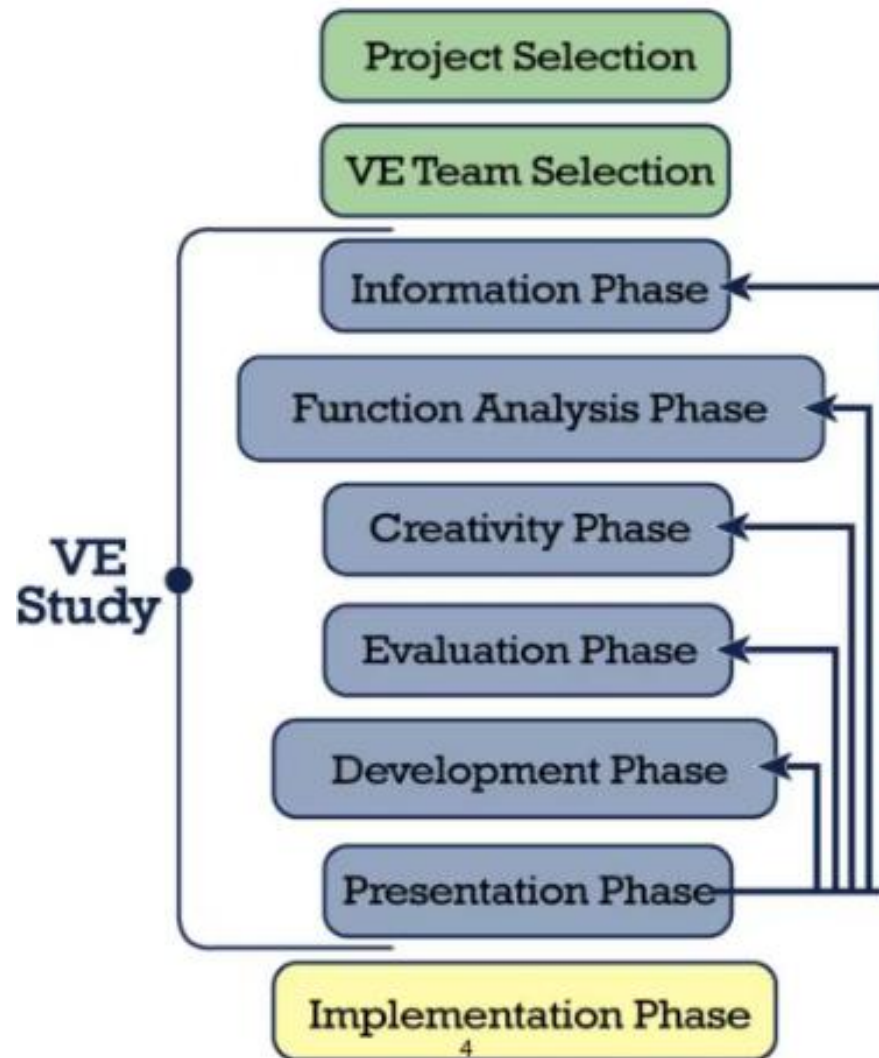


S A V I N G S
I N N O V A T I O N
B A L A N C E
I N G E N U I T Y
T E A M W O R K



Value Engineering

Value Engineering Study:





Value Engineering

Some of the commonly used project materials which can be value analyzed are:

- Raw and semi-processed materials
- Parts, components, sub-assemblies
- Sub-contracted materials, parts, components etc.
- Finishing items (tiles, paints, polishes etc.)
- Maintenance, repair and operation items
- Transportation and material handling processes
- Printing and stationary items
- Regular consumption items
- Capital good/ plants, machinery, equipment, tools
- Utilities e.g. power and water supply



Selection of Materials to be Value Analyzed

- Select materials of high usage value **(A)** and medium usage value **(B)** categories. Even a **small percentage of reduction**, after **value analysis**, substantial reduction in overall cost.
- Select the material whose **quantity, prices and deliveries** are **not favorable**. **Value analysis** can aid in finding substitutes from other sources.
- Select the materials being imported. **Value analysis** can assist in locating **domestic substitutes**



Creating Value Analysis Awareness

Value analysis awareness for material procurement order can be done by using following checklist:

- If the function doesn't add any thing desirable **eliminate** it
- Explore better and cheaper **alternate**
- If the component performing function is a non standard, locate the **standard** item
- **Modification:** If the cost of the item is not reasonable
- **Consumption reduction:** If the consumption of the item can be cut down by value analysis.
- Redesigning components
- Make Decision
- Supply Source
- Imported Components